

Ahmed Ezz El-Deen Mahmoud

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Professional Summary

Systems Engineer with production ownership of ADAS features (Adaptive Cruise Control – ACC) across Valeo platform and multiple OEM programs. Expertise in system requirements definition, requirements traceability, SiL/HiL/ViL validation and on-vehicle testing for L2/L2+ capable features. Delivered ACC to SOP for Volvo/Geely, BYD, Renault, Volkswagen, Ford and GWM; reduced false-negative target-selection rate by 50%. Practical familiarity with ISO 26262 and SOTIF practices and experienced in cross-domain integration with software, hardware and sensor teams.

Key Skills

Systems Engineering, Requirements Engineering, ADAS, Adaptive Cruise Control (ACC), L2/L2+, System Validation (SiL / HiL / ViL), Vehicle Integration, Requirements Traceability (DOORS, JIRA), Sensor Fusion (camera, radar, LiDAR), Target Selection & Tracking, Ego-motion / Path Prediction, Test Specification, Euro NCAP, ISO 26262, SOTIF, Root-Cause Analysis, C/C++, Python, Git, Jenkins, MATLAB/Simulink, ROS, CAN diagnostic tools

Professional Experience

ADAS Feature Owner – Systems Engineer

Valeo, Cairo, Egypt

Mar 2025 – Present

Lead system engineering and end-to-end feature ownership for Adaptive Cruise Control (ACC) Valeo platform across multiple global OEM programs (**Volvo/Geely, BYD, Renault, Volkswagen, Ford, GWM**).

- **System Engineering & Feature Ownership (Adaptive Cruise Control - ACC):** Defined system requirements for L2 / L2+ capable features, acceptance criteria and maintained requirements traceability (DOORS / JIRA) across multiple Valeo platforms (**EU / CN**).
- **Requirements Translation & System Architecture:** Translated customer and regulatory requirements into functional specifications and use cases, shaped system architecture and interface requirements with cross-functional teams (software, hardware, sensors) for production-ready implementations.
- **Validation & Test Management:** Developed system-level test plans and specifications; managed validation campaigns across SiL / HiL and on-vehicle testing, including **Euro NCAP** scenarios; authored test reports for production sign-off.
- **On-Vehicle Testing & Root Cause Analysis:** [Owning B1 Driving license from BMW](#), Conducted hands-on on-vehicle testing and field validation; performed root-cause analysis for complex system/performance faults (sensor calibration, CAN/comm, perception/decision mismatches, control anomalies); coordinated cross-functional corrective actions.
- **Algorithm Refinement & Safety Integration:** Improved ACC robustness and longitudinal control through algorithm refinement (target selection, tracking, control logic), integrated safety

considerations using ISO 26262 / SOTIF practices; contributed **patented** improvements in ego-path prediction / target-selection logic.

ADAS Software/ System Engineer

Valeo, Cairo, Egypt

Aug 2022 – Mar 2025

Focused on ACC algorithm design, decision-making logic and validation for high-speed ADAS functions.

- **Enhanced Algorithms for Adaptive Cruise Control (ACC):** Developed and implemented **patented** decision-making and target-selection algorithms for ACC, significantly improving target tracking robustness across diverse traffic scenarios (vehicles, cyclists, motorbikes, scooters, trucks, pedestrians, construction zone objects). Achieved more than **50% reduction in false-negative (FN)** target selection rates through innovative algorithm enhancements.
- **ACC Target selection System/ Software Ownership & Troubleshooting:** owned first-line root-cause analysis and cross-functional troubleshooting for ACC target selection, including complex scenarios like **roundabouts**. Drove algorithm and integration improvements to consistently meet SOP quality targets.
- **Production Release Management:** Successfully delivered production releases for multiple OEM programs, managing requirements, testing, and comprehensive documentation.
- **ADAS Validation & Testing:** Contributed to SiL validation and authored system-level test cases and scenario simulations to verify high-speed ADAS performance against system requirements. Provided critical support for HiL and on-vehicle testing, ensuring production readiness.
- **ADAS Features Development & Integration:** Contributed to the development of key ADAS features including lane-centering (**LCS**), lane-change assist (**LCA**), Blind Spot Detection (**BSD**), Close Vehicle Warning (**CVW**), Rear Collision Warning (**RCW**), and Cross Traffic Warning (**RCTW/FCTW**), integrating decision logic with perception and control pipelines.
- **Ego-Motion & Path Prediction:** Designed and refined ego-motion state estimation and ego-path prediction algorithms to support ACC, enabling reliable real-time trajectory prediction for longitudinal control.

Patents

- [DE 102024128954A1— Detecting a Lane Change Maneuver & Differentiating Ego Driving on Lane Boundary \(Issued Sep 9, 2026\)](#)
- [DE 102023118015A1— Filtered Boosting of Yaw Rate Signal for Enhanced Ego-Vehicle Path Prediction \(Issued Jan 9, 2025\)](#)
- [CN 202510993022 — ACC Target Selection for Crossing Objects \(Filed Jul 18, 2025\)](#)

Education

B.Sc. Mechanical Engineering, Mechatronics Major. Ain Shams University in Cairo (2022)